

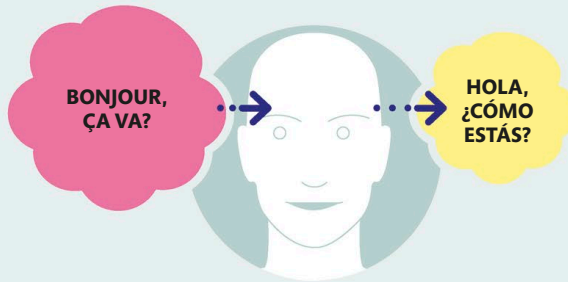


Interfering memories

Our brains experience interference, particularly when information is similar. Learning new information can block recall for old, and old information can also affect new. These problems might arise because the wrong memory trace is activated when you go to recall the information, blocking access to the right one. Or it may be that old information can disrupt consolidation of new, and if successful, the new memory may actually replace the old one.

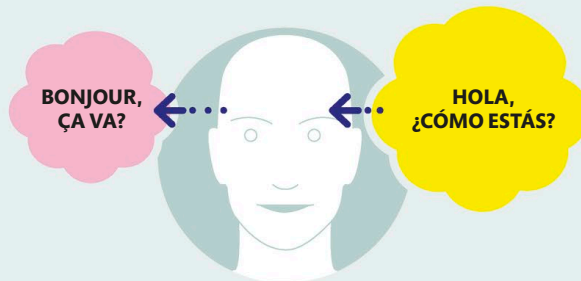
Proactive interference

Old memories may disrupt new ones. For example, when starting to learn Spanish, you may experience interference from French words learned as a child.



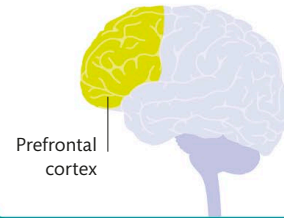
Retroactive interference

If you later went to speak French and instead spoke Spanish, that would be new memories disrupting the recall of old ones.

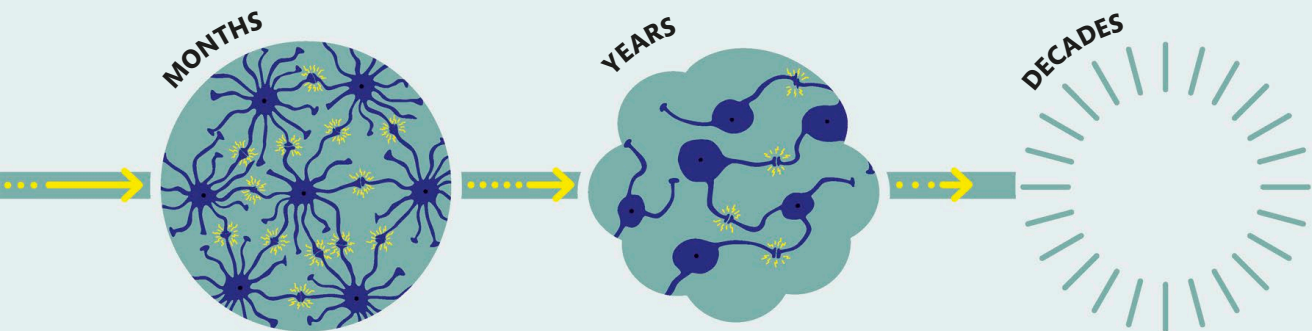
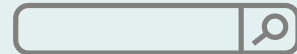


ACTIVE FORGETTING

Forgetting seems passive, but you can choose to forget. In one study, subjects' prefrontal cortices—involved in suppression—were activated when they were told to forget a specific word.



WE MAY BE LESS
LIKELY TO RECALL
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ONLINE; THIS IS THE
GOOGLE EFFECT



1 Storage

Long-term memories are stored in the cortex as networks of connections. These form and strengthen over weeks or months. Recalling a memory activates it, strengthening the synapses and making the memory easier to retrieve later.

2 Memory fades

If months or years pass before you recall a memory, it may begin to fade. Without reactivation, connections between nerve cells are not strengthened. Specific details about special events, such as the food you ate at your wedding, may be forgotten.

3 Losing a memory

One theory for forgetting is that synapses that are not in use become weaker and are eventually pruned away, taking that memory with them. The longer a memory is inactive, the more likely it is to be lost through this process.